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Vegetation Changes in a Shallow African Lake: Response of the Vegetation to a Recent Dry Period

by

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ABSTRACT

This paper shows how the bed of Lake Chilwa in Malawi was colonised by plants during a recent dry period. *Aeschynomene pfundii* and *Diplachne fusca* were the principal colonising species on the alkaline mud flats. The subsequent changes in the vegetation after the lake refilled are described.

Aeschynomene pfundii survived the flooded conditions for four years and eventually died out as seed germination could not occur underwater. The salinity of the lake during the low water and drying period prevented the lake bed from being extensively colonised by *Typha domingensis*.

INTRODUCTION

It is widely recognised that water regime, including water depth and fluctuations in water level, is the most important factor controlling the distribution and species composition of aquatic and littoral plant communities (HEJNY 1971, KVET 1971, SEGAL 1971). Because changes in the controlling factor can be both rapid and substantial, wetland and aquatic communities make ideal sites for the study of vegetation dynamics. In spite of this it is only very recently that studies have been made of the short-term dynamics of wetland vegetation, and SCULTHORPE (1967) points out that data of this type are especially lacking for the tropics.

A number of studies are reported for temperate water bodies on the effects of drawdowns and dry periods on the vegetation (KADLEC 1962, HARRIS & MARSHALL 1963, DESELM & SHANKS 1967,

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KUFLIKOWSKI 1968 etc.). In tropical Africa, MITCHELL (1969) and MAGADZA (1970) have pointed out the importance of large water level fluctuations to the aquatic vegetation of Lake Kariba.

Large-scale water level fluctuations, frequently resulting in complete drying are a common feature of East African shallow lakes (BROWN 1965, VESEY-FITZGERALD 1970, GUNN 1973, RICHARDSON & RICHARDSON 1972). GUNN (1973) suggests that the periodic drying up of Lake Rukwa in Tanzania is a cyclic event occurring three to four times per century.

Lake Chilwa in Malawi appears to be similar. The lake dried up in 1882 (O'NEILL 1884), 1920, 1934 (GARSON & CAMPBELL-SMITH 1955) and 1968. This recent dry phase in Lake Chilwa has been the subject of a number of investigations (MORGAN 1968, MCLACHLAN & MCLACHLAN 1969, MOSS & MOSS 1969, MORGAN & KALK 1970). This paper outlines the effects of the 1968 dry period on the vegetation of Lake Chilwa with special emphasis on the changes occurring when the lake refilled.

THE STUDY AREA

The environment

Lake Chilwa lies on an alluvial plain about 30 km east of the Rift Valley. The lake is shallow (average water depth 2 m) and has an open water surface area of 700 km^2. It is completely surrounded by large tracts of swamp. The water conductivity and pH are relatively high because the lake is endorheic (HOWARD-WILLIAMS 1972, MCLACHLAN et al. 1972). The soils are generally considered to be alkaline clays (MCLACHLAN & MCLACHLAN 1969) although there are a few sandy patches. A distinct gradient in soil characteristics from the swamps into the lake was shown by HOWARD-WILLIAMS (1972).

Lake Chilwa has a radial drainage pattern, although the perennial and largest rivers are all in the southern half of the catchment, especially in the south west (Fig. 1). Here also, the swamps are narrower than in other parts of the lake. These features had an important influence on the course of vegetation development when the lake dried, as will be discussed shortly.

The vegetation

HOWARD-WILLIAMS (1973) and HOWARD-WILLIAMS & WALKER (1974) classified the vegetation of Lake Chilwa into six important types; floodplain grassland, neutral to acid marsh, alkaline marsh, swamp transition, swamp, and disturbed vegetation.

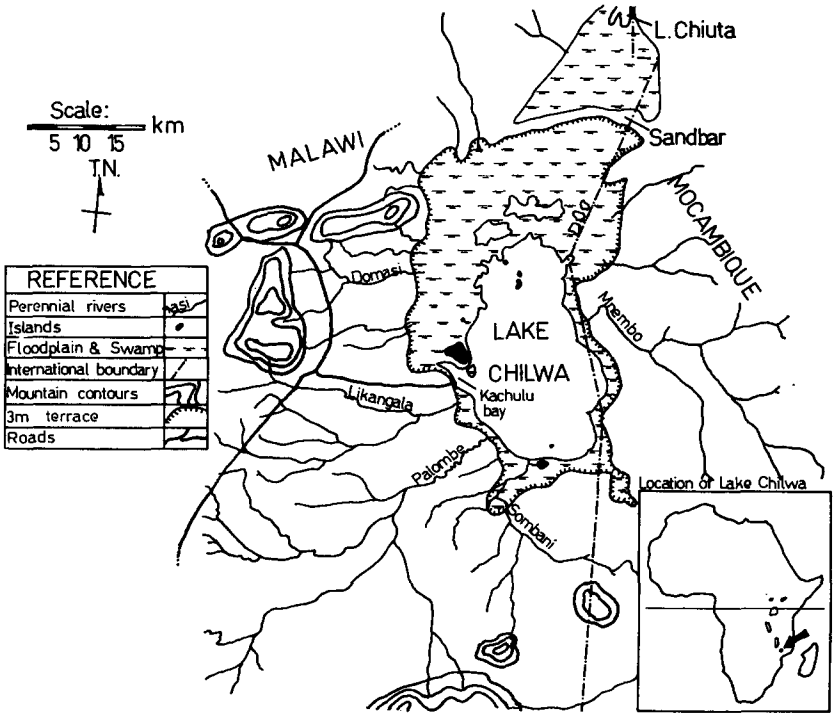

Fig. 1. Lake Chilwa basin showing the lake, drainage pattern and surrounding swamp and floodplain.

Floodplain grassland is found in areas which are dry for more than nine months per year. *Hyparrhenia rufa* (NEES) STAPF, *Cynodon dactylon* (L.) PERS., *Sporobolus pyramidalis* BEAUV. and *Eragrostis gangetica* (ROXB.) STEUD. are the predominant species.

Marsh vegetation characterised by sedges occurs in areas that are dry for only half the year. *Cyperus procerus* ROTTB., *Cyperus articulatus* L. and *Scirpus maritimus* L. are common everywhere, whilst *Cyperus papyrus* L. is confined to river mouths. The grass species most common in these moister regions are *Leersia hexandra* SWARTZ, *Vossia cuspidata* GRIFF and *Echinochloa pyramidalis* (LAM.) HITCHC. & CHASE. In the alkaline marshes *Cyperus* spp., *Scirpus maritimus* and *Fiurena ciliaris* (L.) ROXB. are common sedges. The large legume *Aeschynomene pfundii* TAUB. is found in clumps in the marsh on the western and southern sides of the lake.

Swamp transition vegetation occurs mostly in the north of the lake on alkaline soils, which, although they have no standing water

for much of the year, are kept moist by wind generated water surges from the lake which occasionally flood the area. The vegetation is made up of the grasses *Diplachne fusca* (L.) BEAUV., *Panicum repens* L. and *Vossia cuspidata* together with patches of the sedges *Cyperus laevigatus* L. and *Scirpus maritimus*. Scattered *Typha domingensis* PERS. clones may also occur.

By far the largest area of vegetation is swamp, dominated by *Typha domingensis*, occurring largely as pure stands. Swamp transition and marsh species may however occur in the *Typha* on the landward edges, whilst several free floating species such as *Pistia stratiotes* L., *Ceratophyllum demersum* L. and *Utricularia* spp. are found on the lake edges of the swamp. Out in the lake itself, apart from a few clumps of *Typha*, the large sedge *Scirpus littoralis* SCHRAD., and the aquatic grass *Paspalidium geminatum* (FORSK.) STAPF occur in patches. These appear to be the only plants able to live in the relatively deep water and to tolerate heavy wind and wave action. There is no submerged vegetation due to the high water turbidity.

THE DRY PERIOD

Fig. 2a gives the mean annual lake level, and discharge into Lake Chilwa of one of the major rivers for the period before, during, and after the dry period of 1968. The river flow figures indicate that the lake dried up as a result of decreasing rainfall in the whole catchment before 1968. In the 1969 rainy season there was a spectacular rise in lake level when the lake refilled in a few months. Fig. 2b shows the changes in major ion concentrations in the water, indicated by changes in the electrical conductivity, during the same period. The maximum measured conductivity was 12 000 μmho cm^{-1}. Although the conductivity probably increased beyond this figure as the lake receded to its lowest level when only a few hundred square meters of water were left (Moss & Moss 1969), it was not possible to reach the water at this time.

The pattern of the data in Fig. 2 is such that the changes associated with the dry period can be conveniently divided into four distinct phases; the pre-drying phase, the dry phase in 1967—1968, the refilling phase in 1969, and the post-filling phase from 1970 on. The sequence of vegetation change will be described in each of these phases.

The pre-drying phase

According to Fisheries Department personnel who were present in the Chilwa area before the lake dried, the margins of the open

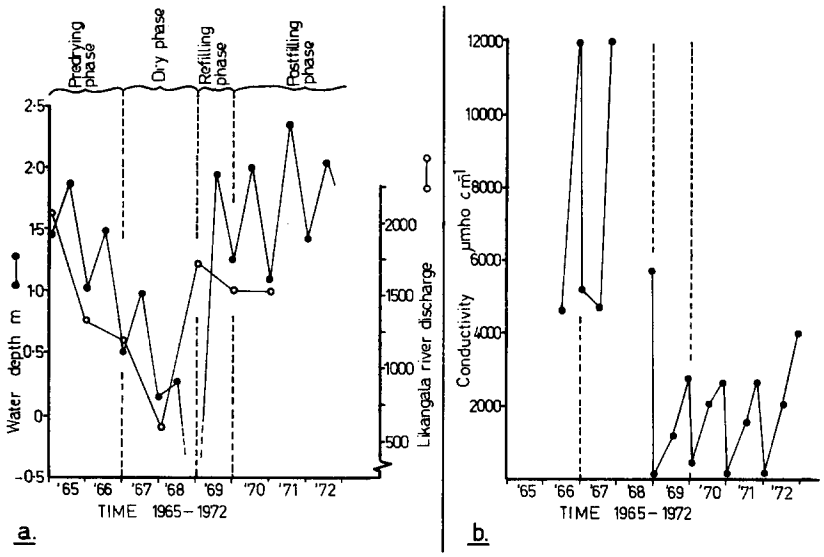

Fig. 2. Changes in water depth, inflow, and conductivity before, during and after the dry period in 1968. The duration of the various phases during this period – described in text – are also shown. Figures given for Likangala River discharge are relative. To obtain total m^3 yr^{-1} multiply by 86400.

lake were defined almost entirely by *Typha domingensis* (R. KIRK, C. RATCLIFF, personal communications). The lake margin before 1967 can therefore be represented as shown in Fig. 3a.

The dry phase

By 1967 the lake waters had receded from the swamp margin and by mid 1968 there was only a relatively small pool of water remaining in the south-central region of the lake (MORGAN & KALK 1970). The enormous area of exposed drying mud which the lake waters had once covered provided a niche available for colonisation by adventive plants from the marginal vegetation. Seeds of these plants were either wind blown on to the mud, or were washed on to the mud by incoming river water during the rainy seasons of 1967—1968 and 1968—1969. In addition there were seeds already present on the lake bed which were able to germinate after exposure to air.

However, with the exception of the sandy areas, particularly that in the south-west of the lake, the dry mud proved to be a very inhospitable environment because of a number of factors. High concentrations of sodium bicarbonate and sodium chloride in the waters resulted in a crust of salt crystals forming on the mud as

drying progressed (Moss & Moss 1969). In addition, the exposed black muds reached high temperatures; over 40°C has been recorded on similar dry muds on the north plains of Lake Chilwa. The mud developed large cracks (McLachlan et al. 1972) which would have had the effect of tearing the roots of small plants. In these conditions only three species survived on the mud flats, *Diplachne fusca*, *Cyperus laevigatus* and *Aeschynomene pfundii*, species characteristic of alkaline clays in the lake basin.

In the few sandy areas, however, and on the mud flats opposite the large rivers in the south-west, environmental conditions were not as harsh. The sandy soils, for instance, have lower salt concentrations and do not form cracks on drying, and the rivers provide fresh water. Consequently, the flora which developed in these regions when the lake dried was much richer than on the alkaline muds, as described below.

The refilling phase

The lake margins were mapped in 1969 by aerial survey and prismatic compass, and are shown in Fig. 3b. Eighty four km^2 of lake bed were colonised by new vegetation, *Diplachne fusca* and *Aeschynomene pfundii* which occupied 48 and 34 km^2 respectively, accounting for most of it. For reasons described above, these species occurred as almost pure stands. However, on the sandy soils and opposite the large river inflows in the south-west of the lake a fairly complex vegetation type with a large number of species developed. A floristic assessment of one such region opposite the Likangala River mouth was carried out regularly beginning in January 1969. At this time the water conductivity was low (Fig. 2) and the water was clear, so that the bottom muds were visible. The newly developed plant community here consisted of a sward of *Diplachne fusca* interspersed with distinct clumps of other aquatic grasses, namely *Paspalidium geminatum*, *Vossia cuspidata* and *Echinochloa pyramidalis*. *Aeschynomene pfundii* and *A. nilotica* Taub. were commonly found growing up through the grasses. The trailing plants *Ipomoea aquatica* Forsk. and *Ludwigia stolonifera* (Guill. & Perr.) Raven spread across the water surface. *Cyperus alopecuroides* Rottb., *C. articulatus*, *Nymphaea lotus* L. and *N. caerulea* Savigny were also common. Because of the clear water, submerged aquatic plants such as *Ottelia exserta* (Ridl.) Dandy, *O. ulvifolia* (Planch.) Walp. and *Nitella* sp. were present. Within a few months the species composition of the newly developed vegetation had changed markedly (Table I) due to four interrelated factors: (i) The water depth increased rapidly from 60 cm in January to over 2 m in April (Fig. 2) as the lake filled. This resulted in the death of the *Cyperus*

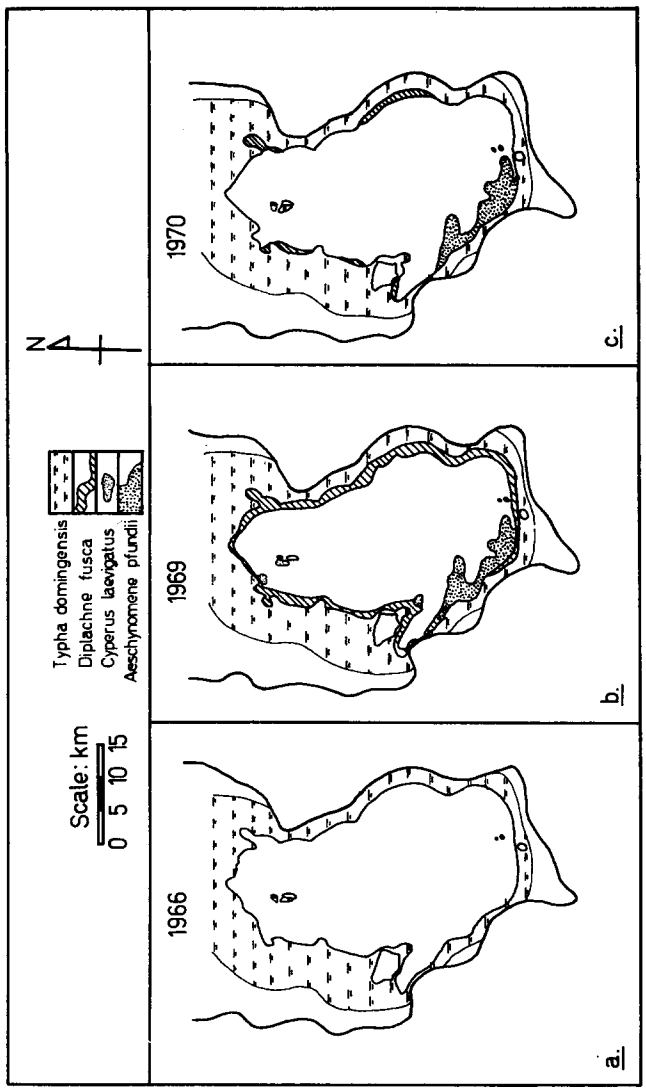

Fig. 3. Changes in the swamp-lake margin resulting from the dry period. a – predrying phase; b – refilling phase; c – early post-filling phase.

species and some of the smaller dicotyledons such as *Alternanthera sessilis* (L.) DC. and *Polygonum limbatum* MEISN. (ii) Wind action mixed the water column disturbing the bottom muds, as described in MCLACHLAN et al. (1972) thereby causing a sharp decrease in water clarity (Secchi disc reading was only 6 cm a few months after refilling), so that the submerged species died. (iii) Water

TABLE I

Changes in species composition of the temporary swamp association during and after the refilling of Lake Chilwa.

Species	Feb '69	Feb '70	Feb '71	Feb '72	Feb '73
Aeschynomene pfundii TAUB.	+	+	+	+	
Nymphaea lotus L.	+	+	+	+	
Paspalidium geminatum (FORSK.)STAPF	+	+	+	+	
Diplachne fusca (L.)BEAUV.	+	+	+		
Nymphaea caerulea SAVIGNY	+	+	+		
Ipomoea aquatica FORSK.	+	+	+		
Vossia cuspidata GRIFF.	+	+	+		
Spirodela polyrhiza (L.)SCHLEID.	+	+			
Ceratophyllum demersum L.	+	+			
Pistia stratiotes L.	+	+			
Echinochloa pyramidalis (LAM.)HUTCH.	+				
Aeschynomene nilotica TAUB.	+				
Cyperus laevigatus L.	+				
Cyperus alopecuroides ROTTB.	+				
Cyperus digitatus ROXB.	+				
Ludwigia leptocarpa (NUTT.)HARA	+				
Ludwigia stolonifera (GUILL.&PERR.) RAVEN	+				
Ottelia exserta (RIDL.)DANDY	+				
Ottelia ulvifolia (PLANCH.)WALP.	+				
Alternanthera sessilis (L.)DC.	+				
Nitella sp.	+				
Polygonum limbatum MEISN.	+				

conductivity increased from 200 μmho cm^{-1} in January to almost 1 500 μmho cm^{-1} in June 1969 (Fig. 2). (iv) Prolonged wave action reduced the community further, the trailing *Ludwigia* spp. being particularly susceptible.

By February 1970, few of the original species remained (Table I).

The post-filling phase

Cyperus laevigatus, being a relatively short plant, was rapidly covered by the rising waters and within a few months had disappeared.

Within one year of refilling the *Diplachne fusca* began to die as a result of the prolonged deep water conditions (Fig. 3). This species is known to be susceptible to long periods of flooding (VESEY-

FITZGERALD 1970). By February 1972 there was no *D. fusca* left out in the lake, and most of the *Aeschynomene pfundii* was dead. By 1973 the colonising vegetation had disappeared completely and the margins of the lake were once again made up purely of *Typha* as in the predrying phase.

FACTORS AFFECTING COLONISATION OF THE DRY LAKE BED

When considering the colonisation of the dry lake bed of Lake Chilwa by plants, several peculiarities become apparent. Fig. 3b shows that *Diplachne fusca* was found in a zone all around the lake, and *Aeschynomene pfundii* occurred only in the south-west. Table I indicates that of the colonising species, *Aeschynomene pfundii* was the most tolerant of the flooded conditions when Lake Chilwa refilled. During its period of growth in Lake Chilwa large numbers of seeds were produced, and yet no sexual regeneration occurred. Furthermore, in spite of an enormous reproductive capacity there was no massive colonisation of the dry lake bed by the surrounding *Typha* populations such as was recorded in a dry temperate lake bed in North America by HARRIS & MARSHALL (1963). Aspects of the early growth characteristics of *Aeschynomene pfundii* and *Typha domingensis* were investigated experimentally to try to explain these apparent anomalies. Experiments on these two species are described separately below.

Aeschynomene pfundii

Aeschynomene pfundii is a herbaceous legume growing to a height of 4 m, with thick trunks made up of soft, thin-walled vascular tissue containing much xylem parenchyma. In its natural habitat on seasonally flooded soils it grows in distinct clumps, a distribution pattern caused by heavy seeds 0.5 to 1.0 cm long which cannot be easily dispersed. On the south-west shore of Lake Chilwa the swamps are relatively narrow and, as pointed out earlier, there is a large water inflow here. This region is therefore the only place where the seeds of *Aeschynomene pfundii* could be washed easily through the swamp into the lake. This factor probably accounts for the peculiar distribution pattern shown in Fig. 3.

Attempts were initially made to germinate the seeds in deionised water. There was no germination after six months and the seeds were still extremely hard. Several methods of dormancy breaking, such as scratching and burning the testa and soaking the seeds in acid were tried. It was eventually found that maximum germination was obtained after soaking the seeds for one hour in concentrated

TABLE II

Composition of the culture solution

	mmol dm^{-3}
KNO_3	0.742
$Ca_3(PO_4)_2$	0.083
$CaSO_4 \cdot 2H_2O$	0.145
$MgSO_4 \cdot 7H_2O$	0.102
$FeCl_3$	0.006
NaCl	0.411 to 20.548
Na_2CO_3	0.166 to 8.449

sulphuric acid. The fact that the testa is so hard means that the seeds probably lie dormant for many years in the field before microbial activity softens the testa sufficiently to allow germination.

An examination of the environmental factors operating on the seasonally flooded soils where *A. pfundii* naturally occurs and those in the lake, indicated three possible reasons why *A. pfundii* does not occur in deep water, and why it did not regenerate after initial colonisation of the lake bed in the dry phase. First, almost no light penetrates to the bottom muds due to the turbidity of the open lake water; second, the salt concentrations in the waters and muds of the lake are higher than on land; third, the continually submerged lake muds have a high oxygen demand (MORGAN 1972) in flooded conditions. These possibilities were evaluated experimentally, using seeds pre-treated with sulphuric acid.

The effect of darkness on germination

Four replicate containers each contained 20 *A. pfundii* seeds on moist filter paper; two containers were kept in the light (fluorescent white light, 2 000 lux) and two in the dark at 25°C. After three days there was 90 and 92.5% germination in the light and dark respectively. There was no further germination. The dark conditions prevailing on the flooded lake bottom would therefore have no effect on germination.

The effect of salinity on germination and early growth

A culture solution prepared as shown in Table II was used. The levels and proportions of the major ions approximate those

prevailing in Lake Chilwa. McLachlan et al. (1972) showed that as the lake dried and the salt concentration of the water increased, the relative importance of Na^+, Cl^-, and $CO_3^{--} + HCO_3^-$ increased with increasing conductivity. NaCl and Na_2CO_3 were therefore the salts used to vary the salinity of the culture solution. The treatments contained equal amounts of NaCl and Na_2CO_3 in the following concentrations: 0.87, 10.87, 21.75, 32.62, and 43.5 mmol dm^{-3} of Na^+.

One hundred *A. pfundii* seeds were placed on plastic gauze screens stretched over the surface of 1.5 dm^3 pots containing the culture solutions. The pots were kept under continuous fluorescent white light illumination (2 000 lux) at 25°C. Forty seeds were allocated to each treatment. There was no significant difference in germination between the treatments after three weeks.

Fifty seeds were then germinated on the gauze frames in the lowest sodium treatment, and after a week they were divided equally among the five salinity treatments. After two-and-a-half weeks the plants were dried and weighed. The results are shown in Table III.

TABLE III

Dry masses of Aeschynomene pfundii seedlings grown in different salinities.

Treatment	mmol dm^{-3} Na^+				
	0.87	10.87	21.75	32.62	43.50
Mean dry mass mg (n=10)	47.6	43.0	36.5	39.5	32.5

Analysis of variance:

Calculated variance ratio = 3.573 ($P = < 0.05$)

Least significant difference between treatments ($P=0.05$) = 12.69 mg

Although mean growth generally delined with increasing sodium concentration, only the growth in the highest treatment differed significantly from that in the control (0.87 mmol dm^{-3}) and all the plants were healthy at the end of the experiment. As the highest treatment had NaCl and Na_2CO_3 concentrations higher than have been recorded in the lake since the dry period, it appears that salinity during the postfilling phase would not have affected either the germination or early development of *A. pfundii*.

The effect of submergence and drying on germination and early growth

Equal quantities of moist marsh mud from the south west shore of Lake Chilwa were placed in polythene containers, and deionised

water was added to give depths of 0 (moist mud), 5, 10, 20, and 30 cm. Forty seeds were distributed in four replicate containers per depth treatment and the pots were randomised under continuous illumination (approximately 1 000 lux) at 25°C.

After 10 days there was 80% germination in the 0 treatments in which the seeds were lying on the moist mud in contact with the air. There was no germination in any of the other treatments. This was still the case after 30 days, so water was decanted from two replicates of each of the other treatments. Within 10 days, germination was recorded in all the newly dried pots (mean 47%, range 10 to 80%). There was still no germination in any of the flooded pots. The seedlings in the emptied pots were left to grow for a further 10 days, and the pots were then refilled to their original levels. In the 5 cm treatment the young plants were above the water level, but were covered in the other treatments. The experiment was terminated one month later, when all the seedlings in the 5 cm treatment which were above the water after reflooding were still healthy. All the other seedlings were dead. The seeds in the pots which had been continually flooded had still not germinated.

When related to field conditions, these results indicate that *A. pfundii* seeds will not germinate out in Lake Chilwa unless the lake dries up. After germination, seedling growth must be such that the young plants are kept above the rising lake waters or they will die. *A. pfundii* seeds probably remain viable for many years. The plants recently growing in the lake produced seeds, and more are washed in each year with the rivers. Many of these will almost certainly germinate when Lake Chilwa dries again, and the growth of *A. pfundii* in the lake will be repeated.

Typha domingensis

Over 600 km^2 of the littoral zone of Lake Chilwa are occupied by *Typha*, and there are between four and eight inflorescences per square metre during flowering. Each inflorescence produces about 600 000 seeds (Howard-Williams 1973). Towards the end of each dry season (August to October) when the inflorescences ripen, large clouds of seeds may be seen blowing around the lake and swamps. The germination of these seeds is consistently over 80% in deionised water. It was of interest to know, therefore, why there was no massive colonisation of the open lake by *Typha* when the lake dried. Laboratory observations showed that *Typha* seedlings establish themselves rapidly on a moist mud surface. It was mentioned earlier, however, that out in the lake the salinity of the water was very high when the lake was drying, and that crystalline salt deposits formed on the mud surface. The effect of salinity on the

germination and early growth of *T. domingensis* was therefore investigated.

Nutrient solutions were prepared as shown in Table II. The salinity was again varied by altering the amounts of Na_2CO_3 and NaCl to give the following treatments: 0.87 (control), 21.5, 43.5, 108.7, and 217.5 mmol dm^{-3} of sodium. The highest sodium level was approximately that of the water when the lake was almost dry in 1967. Ten polythene pots, with ten seeds per pot, were allocated to each salinity treatment. The pots were randomised under continuous fluorescent white light illumination (2 000 lux) at 25°C. The results are shown in Table IV.

TABLE IV

Germination of Typha domingensis in different salinities after 12 days.

Treatment	mmol dm^{-3} Na^+				
	0.87	21.75	43.5	108.7	217.5
Mean % germination (n=100)	82	64	54	22	6

Analysis of variance:

Calculated variance ratio = 16.9 ($P = < 0.001$)

There was no further germination after the end of the experiment. Germination was therefore strongly inhibited by the high concentrations of NaCl and Na_2CO_3.

The root length, taken as an indication of seedling development, was measured in each pot after 20 days. The results are given in Table V.

Root growth was markedly inhibited above 43.5 mmol dm^{-3} Na^+ and most seedlings in the two highest Na^+ treatments were dead at the end of the experiment. There was no significant difference between root growth in the three lowest treatments, however, even when analysed separately.

These results indicate that both germination and early growth are almost completely inhibited in the high NaCl and Na_2CO_3 concentrations which occur when the lake dries out. These conditions would therefore have prevented the spread of *Typha* on to the dry lake bed. Seedling growth is not affected by 43.5 mmol dm^{-3} of Na^+ which is higher than has normally been found since the lake refilled. Salinities in a normal year therefore do not affect early growth of *Typha*.

TABLE V

Seedling root growth of *Typha* in different salinities.

Treatment	mmol dm^{-3} Na^+ 0.87	21.75	43.5	108.7	217.5
Mean root length cm (n=100)	2.34	2.33	1.96	0.38	0.15

Analysis of variance:

(i) All treatments: calculated variance ratio = 15.69 (P= <0.001)
Least significant difference between treatments (P=0.05)
= 0.872 cm

(ii) The three lowest treatments: calculated variance ratio = 11.20 (P= <0.001)
Least significant difference between treatments (P=0.05)
≅ 0.744 cm

Discussion

The development of, and change in, vegetation on a newly dry lake bed always invites considerable comment, firstly because of the rapidity of change and colonisation, and secondly perhaps because the visual change is so striking. However, the process is no different from that which occurs on a newly ploughed field. Seeds already present germinate, and adventive species with large numbers of light seeds quickly establish themselves in the unoccupied areas. Often in lakes which dry out, or are subject to man-induced drawdowns, a further initial change which occurs is the rapid disappearance of communities of floating, floating leaved, and submerged hydrophytes (Kadlec 1962, Harris & Marshall 1963, Kuflikowski 1968). If the lake bed remains dry, secondary successional changes in the colonising vegetation soon become apparent and the initial adventive species are replaced by larger perennial emergent plants such as *Scirpus* and *Typha* (Harris & Marshall 1963, Deselm & Shanks 1967). Reflooding usually results in the rapid disappearance of the adventive annual species and, depending on water depth, the slower but ultimate disappearance of the perennials (Harris & Marshall 1963).

The changes in the vegetation on the lake bed after Lake Chilwa dried out followed a similar pattern. However, the extent of change and colonisation during the dry phase was small compared with that reported for other dry lakes (Harris & Marshall 1963, Deselm & Shanks 1967). This was a result of the inhospitable environmental conditions prevailing at the time. Salinity seems to

be the major factor influencing colonisation of the lake bed. The three main colonising species, *Diplachne fusca*, *Cyperus laevigatus* and *Aeschynomene pfundii* all occur naturally on high conductivity alkaline clays. *Typha domingensis* seedlings, moreover, were not able to develop on the lake bed because of the salinity (Tables IV and V). On the south west shore, where there is the largest input of fresh water, a diverse flora developed, which was rapidly reduced after reflooding (Table I). Of the colonising species, *Aeschynomene pfundii* was the most tolerant to flooded conditions, and took three to four years to disappear completely after the lake refilled. It eventually died out because the seeds could not germinate under water so there was no regeneration. A dry period in a Polish reservoir resulted in the appearance of a number of new species not previously recorded from it. These often covered large areas (KUFLIKOWSKI 1968). This did not happen in Lake Chilwa, again probably because of the salinity which severely limited the number of species able to invade the area. Apart from temporarily affecting the species composition of the surrounding vegetation, the drying up of Lake Chilwa had profound effects on the amount of dead organic matter. Most swamp systems are characterised by an accumulation of organic matter due to anaerobic conditions prevailing in swamp waters. The Lake Chilwa swamps, however, only have a relatively thin layer, often no more than a few centimetres thick, of undecomposed plant material above the mud surface. In part this can be accounted for by the occasional complete drying up of the lake which must result in considerable oxidation of accumulated organic matter. Progressive accumulations of undecomposed material are normally considered to be the underlying process in an autogenic hydrosere succession (e.g. ODUM 1971, SEGAL 1971). The dry periods in Lake Chilwa therefore help to retard successional development.

The fauna of Lake Chilwa disappeared long before the lake dried up completely in 1967 (MCLACHLAN et al. 1972). Within one year of reflooding, it had largely returned (KALK 1970, MCLACHLAN et al. 1972, MORGAN 1972). Within two years, the colonising vegetation had almost disappeared (Table I, Fig. 3), and no new species had invaded the lake. In other words, within two years of what may be considered a major environmental upset, the flora and fauna of the lake had virtually returned to their former states. PRESTON (1969) stated that biological stability lies in the ability to return to a former state, not in the ability to hold tenaciously to ground once taken, or numbers once achieved. The rapid recovery of the biota after the recent dry period may therefore indicate that Lake Chilwa has a high degree of biological stability.

Summary

Like many other East African shallow lakes, Lake Chilwa dries up periodically. The last dry period was in 1968. This paper discusses the colonisation of the dry lake bed by vegetation, and its subsequent survival. With the exception of sandy areas and regions opposite river mouths where a fairly rich flora developed, the environment on the dry lake bed was unfavourable for plant growth, and only three species survived, namely *Diplachne fusca*, *Cyperus laevigatus* and *Aeschynomene pfundii*. *D. fusca* and *C. laevigatus* were unable to tolerate the deeper water after the lake refilled, and died out after one year. *A. pfundii* thrived in the deep water but was unable to regenerate, so died out approximately four years after flooding. Experimental studies showed that this was due to an inability of the seeds to germinate on the lake bottom rather than to the salinity of the lake water. Salinity was, however, shown to be a factor which almost certainly prevented colonisation of the lake bed by the surrounding *Typha domingensis* populations during the dry period. Both germination and early growth of *Typha* were almost completely inhibited in the high NaCl and Na_2CO_3 concentrations which resulted when the lake dried out.

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